

TITLE: Assessing the Homogeneity of Dose Delivered to Blood During Sweeping Cobalt-60 Total Body Irradiation

ABSTRACT (256/300 words)

Purpose: Total body irradiation (TBI) requires the entirety of the patient's body to be irradiated, and treatment methods vary greatly across treatment centres. For many of these techniques, little is known about the heterogeneity of dose delivered to the circulating blood. To assess the dose delivered to the circulating blood volume during Sweeping Cobalt-60 (Co-60) TBI, a full body blood flow simulation was combined with Monte Carlo generated 4D dose distributions.

Methods: A virtual phantom was used to create a Monte Carlo phantom mimicking a Co-60 TBI treatment set-up. Time-resolved Monte Carlo simulations using modified DOSXYZnrc, for both supine and prone patient positions, were completed. The virtual phantom also enabled the creation of a blood flow model which spatio-temporally tracks blood volumes (BVs) through the phantom. The model is composed of 94 vessel segments and 28 compartments. BV trajectories were determined using a combination of one-dimensional blood flow modeling, in-vivo data, and zero-dimensional compartments. The 4D dose distributions were sampled as the BVs moved through the flow model. For a 12 Gy prescription, in 6 fractions with two patient positions (supine and prone) each and 15 sweeps per position per fraction, the dose to the circulating blood was determined for a single fraction and the entire treatment.

Results: The circulating blood received 1.25 ± 0.98 Gy in one fraction and 10.76 ± 2.36 Gy after all fractions.

Conclusion: A whole-body blood flow model was paired with 4D dose distributions to assess the heterogeneity of the dose to the circulating blood achieved by sweeping beam Co-60 TBI.

Innovation/Impact: This study is innovative because it utilizes a novel methodology to assess the heterogeneity of dose delivered to the circulating blood during Total Body Irradiation (TBI). After assessing sweeping Cobalt-60 TBI this method will be used to address lingering concerns regarding the heterogeneity of the dose delivered to circulating blood achieved by emerging sequential techniques such as VMAT TBI.

Key Results: The study utilized a full-body phantom (Figure 1) to provide both the geometry for the blood flow model and the medium for the Monte Carlo simulations. The blood flow model utilizes a combination of 1D and 0D compartments (Figure 2) and was validated against other published works (Figure 3). How dose is accumulated while travelling through vessels is described in Figure 4, and Figure 5 describes the process in the compartments. For Anterior-Posterior sweeping Cobalt-60 beam technique with a prescribed dose of 12 Gy over 6 fractions (15 sweeps delivered in 6.81 minutes per position per fraction), the resulting dose after one fraction and all fractions are presented in Figure 6.

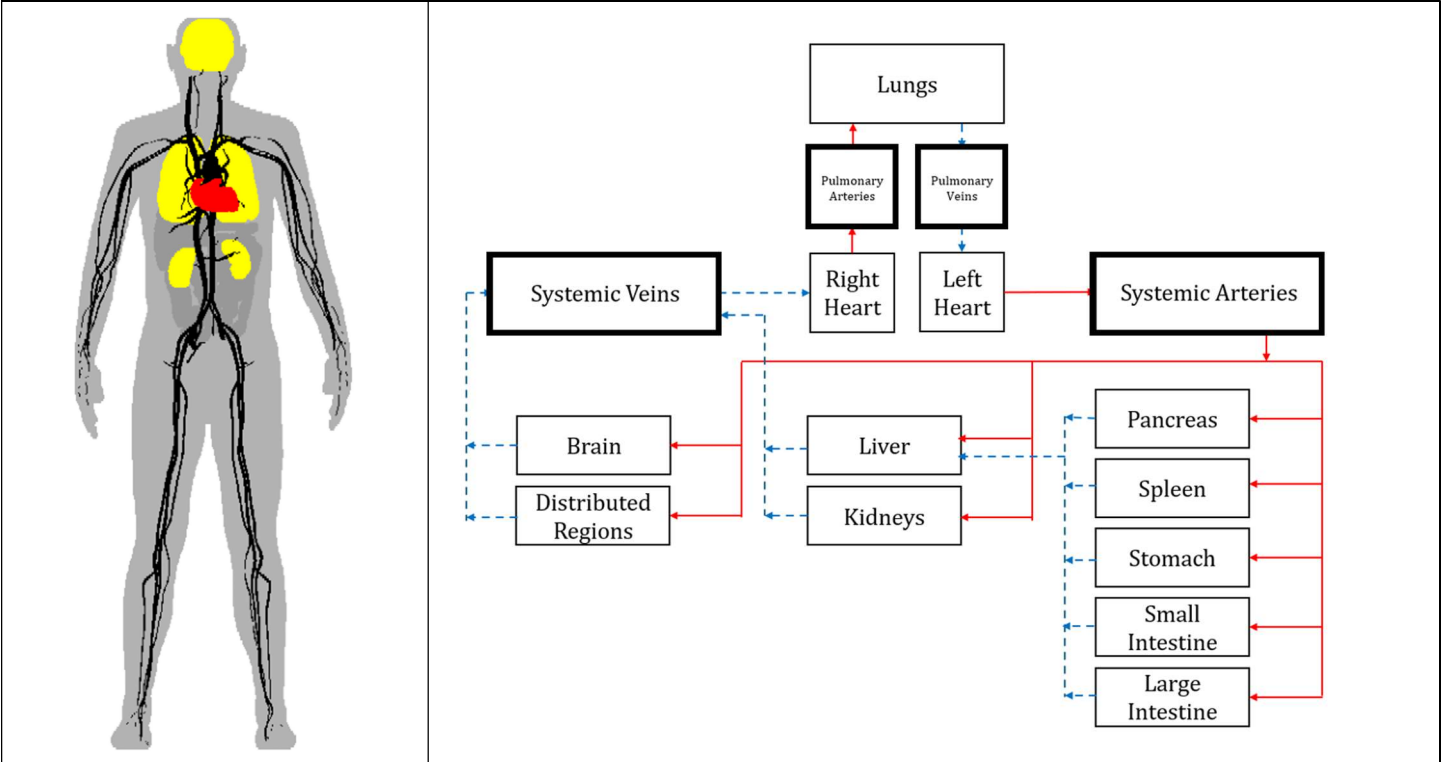
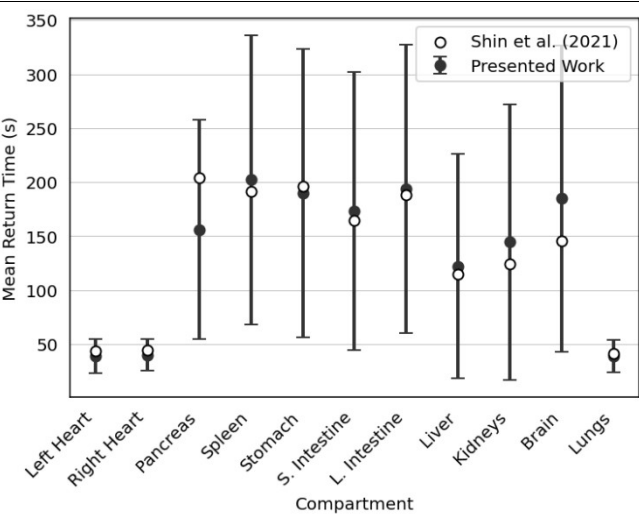


Figure 1: Outline of the full-body phantom with the modelled vessels (black) and a sub-set of the organ compartments visualized (dark grey, yellow and red).

Figure 2: Transition diagram for the blood flow model. Solid lines (red) are arterial flow and dashed lines (blue) are venous. The thick boxes represent one-dimensional vessels, and the thin boxes represent zero-dimensional compartments.

Figure 3: The mean return time (i.e. the time it takes a blood volume to fully cycle through the circulatory system and return to its starting compartment), of our blood flow simulation in comparison to another dosimetry focused blood flow model from Shin et al. [1].



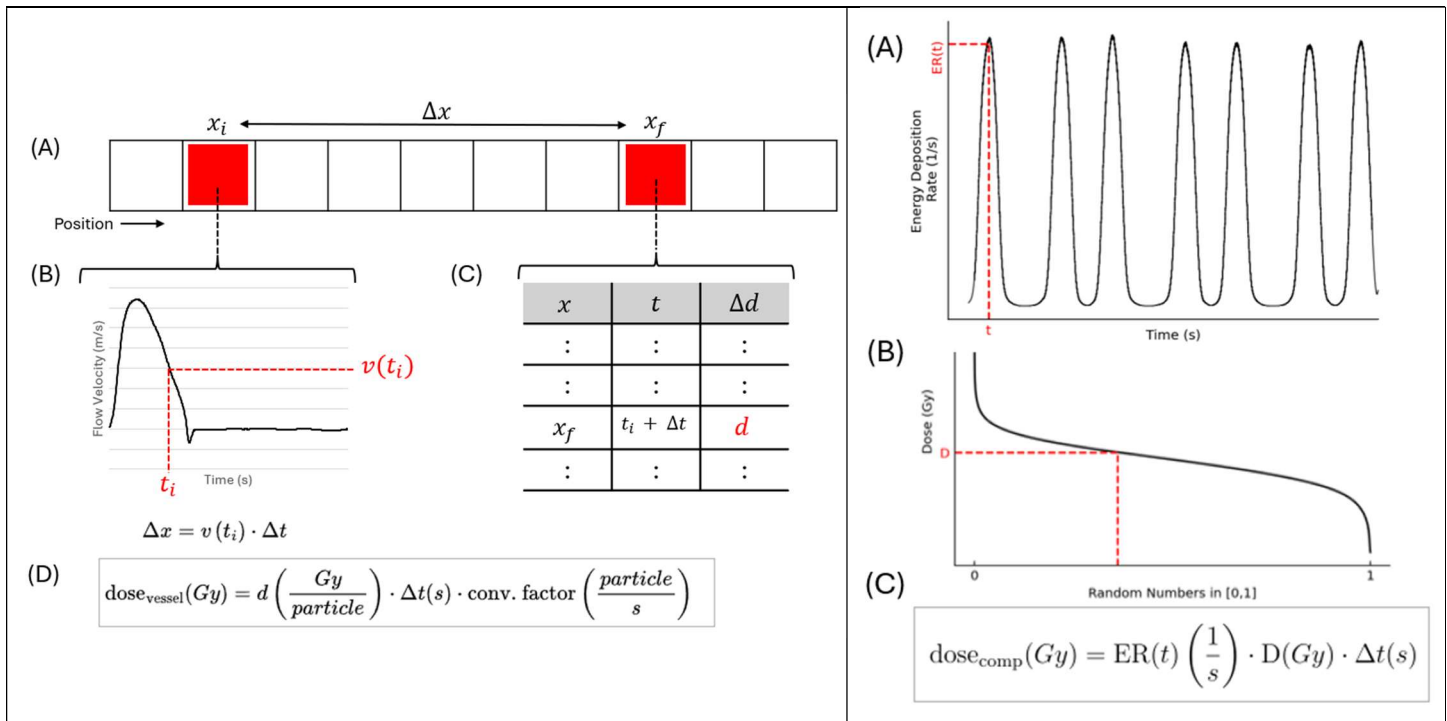


Figure 4: Diagrams explaining how dose to blood volumes (BV) in vessels are calculated over a time step (Δt). (A) A BV's current position (x_i) and the current time (t_i) is used to query the flow velocity, (B), and determine the instantaneous velocity ($v(t_i)$) and thus the BVs change in position (Δx). (C) The new BV position (x_f) and current time is used to query the four-dimensional dose data to determine the dose to the BV (d). (D) Dose is calculated using the queried dose data, time step size and an absolute dosimetry conversion factor.

Figure 5: Diagrams explain how dose to blood volumes (BV) in compartments are calculated over a time step (Δt). (A) The energy deposition rate ($ER(t)$) within the compartment is sampled at the current time (t). (B) The dose value (D) is randomly sampled from the dose volume histogram. (C) Dose is calculated by determining the instantaneous dose rate and multiplying by the time step.

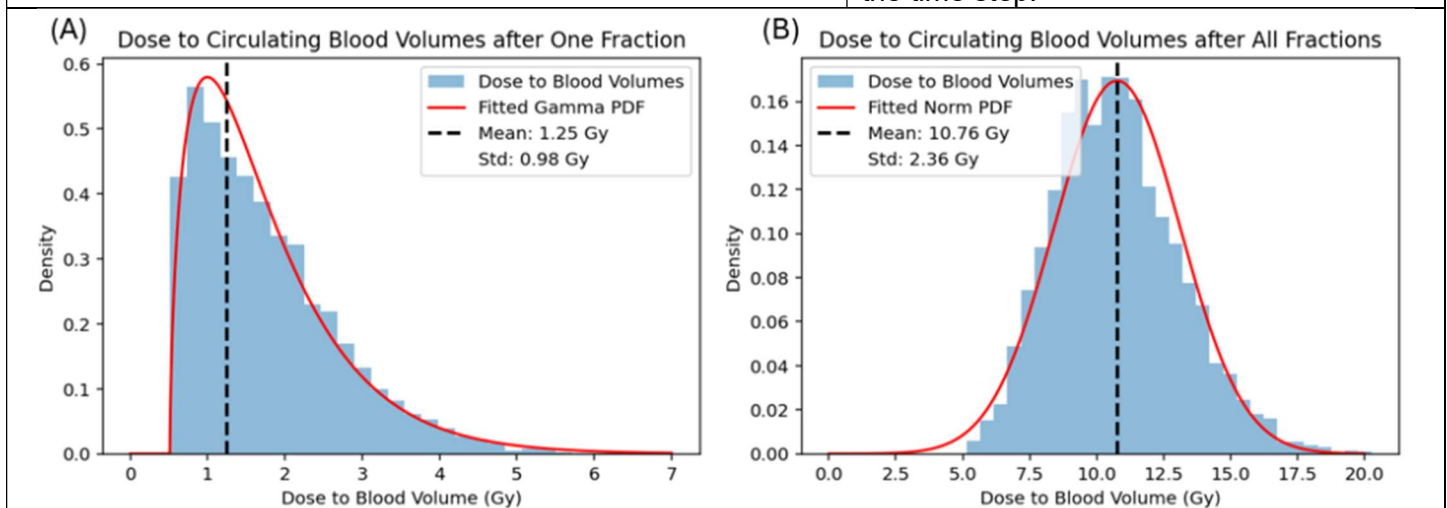


Figure 6: Histograms of the (A) dose to circulating blood volume for a single fraction (B) and the entire treatment plan.

[1] Shin J, Xing S, McCullum L, et al. HEDOS—a computational tool to assess radiation dose to circulating blood cells during external beam radiotherapy based on whole-body blood flow simulations. *Phys Med Biol*.

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